contains an agent that resembles a disease causing microorganism and is often made from weakened or killed forms of the microbe, its toxin or one of its surface protein.

The agent stimulates the body's immune and that it may encounter in the future.

The term vaccine and vaccination were derived from variola vaccine (smallpox of the cow) This term was first discovered by Edward Jenner in 1796.

Types of vaccines:

- Live Attenuated Vaccine
- Inactivated Vaccine
- Subunit Vaccine
- Toxoid Vaccine
- Conjugate Vaccine
- DNA Vaccine
- Recombinant Vector Vaccine



sun is important for the immune system. When a person spends time in the sun their body produces vitamin D.

Live attenuated vaccine:

Live microorganism modified to be less deadly or closely related microorganism that induce immune response. The organism can be attenuated by growing it in prolonged culture. Attenuation means the loss of virulence of the pathogen.

e.g. OPV, MMR (mumps, measles, Rubella) BCG, varicella vaccine, yellow fever.

Inactivated vaccine or killed vaccine:

Whole microorganism destroyed by heat, chemicals, radiation, antibiotics.

e.g. *Hepatitis A vaccine,* Hepatitis B, Pneumococcal polysaccharide, IPV, influenza, Hib, Typhoid.

Toxoid vaccine:

Inactivated toxic compounds is toxoid. [toxins can be inactivated by using formalin]

Toxin +formalin _____ toxoid

e.g. DPT, Antivenom, TT (tetanus toxoid)



- Vaccine prevent more than 2.5 million death every year.
- CDC (Center for Disease Control, USA) has reported a 99% reduction in the incidence of bacterial meningitis caused by Haemophilus influenza.

Subunit vaccine:

A Protein component of the microorganism.

e.g. Surface Protein or Synthetic virus like particles lacking viral genetic material [unable to replicate] e.g. Hepatitis -B

Monovalent Vaccine:

Immunize against single strain of microorganism.

Multivalent Vaccine:

Immunize against multiple antigens strains or micro organism

The children with malnutrition have low resistance to fight against

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